

Draft 1 claimed the first proof that a polygon can be a Spectre. That claim was wrong: Goucher gave one in 2023 [Gou23b]. Draft 2 credits his argument and extends it. The family now contains his polygon.

A TWO-PARAMETER FAMILY OF POLYGONAL SPECTRES

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ABSTRACT. Smith, Myers, Kaplan and Goodman-Strauss showed that the equilateral polygon $\text{Tile}(1, 1)$ is a weakly chiral aperiodic monotile, and that replacing its edges by suitable smooth curves gives the Spectres: shapes that tile the plane, but only non-periodically and only with copies of one handedness, even when reflected copies are allowed. They left open whether piecewise-linear edge paths might permit other tilings. Goucher gave a short argument that one equilateral 28-edge polygon is a Spectre. We extend this to a two-parameter family. Replace each edge of $\text{Tile}(1, 1)$ by a triangular bump or dent, alternating around the tile. The result is a polygon with 27 sides. It is a Spectre for every position of the triangle's peak along the edge and every height below an explicit bound. The family contains Goucher's polygon. In general the sides of our polygons have two lengths, so a corner of one tile could lie inside a side of another. The proof follows Goucher's plan, and is elementary apart from a short computer search for one pair of tiles. Angle sums and side lengths show that the corners of $\text{Tile}(1, 1)$ with angles $2\pi/3$ and $4\pi/3$ meet only each other. This pairs every bump with a dent, and every tiling by the polygon then straightens into a tiling by unreflected copies of $\text{Tile}(1, 1)$.

1. INTRODUCTION

An *aperiodic monotile*, or *einstein*, is a shape whose congruent copies tile the plane, but only non-periodically. The question of whether one exists grew out of Wang's work on tilings by coloured squares [Wan61]. Berger [Ber66] found the first aperiodic set of tiles, and Robinson [Rob71] and Penrose [Pen79] reduced the number of shapes to six and then two. Socolar and Taylor [ST11] found a single hexagonal tile with matching rules, which cannot be realised by shape alone as a connected tile. In 2023 Smith, Myers, Kaplan and Goodman-Strauss [SMKG24a] found the *hat*, a polygon that tiles the plane only non-periodically. For a survey of the path to these tiles, see [Kap24].

Every tiling by hats uses reflected copies of the hat as well as unreflected ones [SMKG24a]. This raised the *vampire einstein* question: is there a shape that tiles aperiodically without its reflection? Smith et al. [SMKG24b] distinguish two forms of the question. A tile is *weakly chiral aperiodic* if it tiles the plane with copies of one handedness, and every such tiling is non-periodic, although tilings that mix both hands may be periodic. A tile is *strictly chiral aperiodic* if it tiles the plane, and every tiling by its congruent copies, reflections allowed, uses copies of one handedness only and is non-periodic.

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The equilateral polygon $\text{Tile}(1, 1)$, a degenerate member of the hat family, is weakly chiral aperiodic [SMKG24b]. It also tiles periodically when both hands are allowed. Smith et al. then replace the edges of $\text{Tile}(1, 1)$ by copies of a curve, oriented so that a tile and its mirror image can no longer meet. The resulting shapes, the *Spectres*, are strictly chiral aperiodic monotiles [SMKG24b, Theorem 2.2], and their tilings have since been studied in detail; see for example [BGMS26]. Their construction uses a smooth curve, and they note that it extends to C^1 curves, which force the vertices of neighbouring tiles to meet. They conclude [SMKG24b, Section 6]:

However, we leave open the question of whether some other paths (e.g., piecewise-linear paths) might permit different tilings, or whether all non-trivial choices of path produce valid Spectres.

A piecewise-linear path adds corners to the tile. A new corner might meet the interior of a side of another tile, and a tiling might then fail to straighten into a tiling by $\text{Tile}(1, 1)$. The first polygonal Spectre that we know of is Goucher’s [Gou23b], posted on 30 May 2023 with a short argument. His polygon is equilateral, with 28 unit sides, and angle sums sort its corners into classes that can meet only each other. This paper shows that triangular paths avoid bad contacts across a two-parameter family, in which the sides have two lengths in general.

The author’s interactive tiling viewer at <https://drbuild.uk/tiles> shows the geometry. It draws tilings by the hat family, including $\text{Tile}(1, 1)$, and by Spectres with several edge shapes. Its Spectre page, with the *Triangle* edge shape in the *Single* arrangement, draws the tiles $X(p, h)$ studied here, with controls for the peak position p and the height h , and marks the height at which the triangles collide.

Main result. Fix $0 < p < 1$ and $h \neq 0$. Replace each edge of $\text{Tile}(1, 1)$ by the two sides of a triangle whose apex lies at signed height h above the point at fraction p along the edge, and give alternate edges the half-turned triangle. The result is the tile $X(p, h)$ of Section 3. Let φ be the angle of the triangle at its apex, and let

$$\delta = \sqrt{6 - 3\sqrt{3}} \approx 0.8966$$

be the least distance between two edges of $\text{Tile}(1, 1)$ with no common vertex.

Theorem 1.1. *Let $0 < p < 1$ and $0 < |h| < \delta/2$. Then $X(p, h)$ is a polygon with 27 sides and a strictly chiral aperiodic monotile. It is a Spectre in the sense of [SMKG24b, Section 2].*

Theorem 1.1 follows from the more general Theorem 5.1, which replaces the bound on $|h|$ by a condition that bumps do not collide. The proof is by hand, except for one pair of tiles, $X(2/7, \pm\sqrt{3}/7)$, where a short exhaustive computer search finishes one step (Proposition 4.4).

Corollary 1.2. *The following polygons are strictly chiral aperiodic monotiles.*

- (1) $X(29/50, 13/50)$, shown in Figures 2 and 4.
- (2) $X(1/2, h)$ for $0 < |h| < \delta/2$. This includes Goucher’s polygon $X(1/2, \sqrt{3}/6)$.

Theorem 1.1 extends Goucher’s example [Gou23b] from one polygon to a two-parameter family. Goucher’s polygon is $X(1/2, \sqrt{3}/6)$ up to scale, and our proof follows the plan of his argument (Remark 7.1). Other authors have stated that straight-edged Spectres exist, without proof. Tatham [Tat23] suggests a jigsaw-style tab or hole on each edge. Kari’s lecture notes [Kar25, Section 7.4] draw straight-edged bumps and dents, in both the half-turn-symmetric and the alternating forms, and state that the resulting tiles have the Spectre properties. Rust [Rus25] states that any consistent piecewise-linear approximation of a curved Spectre works, for example a small orthogonal shift of the midpoint of each edge. Corollary 1.2(2) proves the last construction for small shifts.

Idea of the proof. The polygon $X = X(p, h)$ has two kinds of corner: the 13 corners of $\text{Tile}(1, 1)$, whose angles are multiples of $\pi/6$, and 14 apexes with angles φ and $2\pi - \varphi$. The two sides of each triangle have lengths $a = \sqrt{p^2 + h^2}$ and $b = \sqrt{(1-p)^2 + h^2}$. The alternating arrangement attaches the a -sides to the *blue* corners of $\text{Tile}(1, 1)$, whose angles are multiples of $\pi/2$, and the b -sides to the *red* corners, whose angles are multiples of $2\pi/3$.

We follow the plan of Goucher's argument. First, red corners meet only red corners. For most φ this is an angle count. For the four values of φ where the count allows other contacts, the side lengths rule them out. For one pair of tiles the lengths coincide, and an exhaustive computer search finishes the step. Next, every apex is joined by a b -side to a red corner. The tile across that side at the red corner has a b -side along the same segment, so it has an apex at the same point. The two apexes are of opposite kinds, so each bump fits exactly into one dent. Two tiles of the same hand then meet along whole edges. Two tiles of opposite hands would lay an a -side along a b -side, and following that side leads to a corner where the angles cannot sum to 2π . So every tiling uses one hand, and it straightens into a tiling by unreflected copies of $\text{Tile}(1, 1)$. The results of [SMKG24b] finish the proof.

Organisation. Section 2 fixes terms and collects the results of [SMKG24b] that we use. Section 3 defines $X(p, h)$ and studies its corners. Section 4 shows that red corners meet only red corners and that every bump fits into a dent of the same hand, and Section 5 proves the main theorem. Section 6 treats the examples, and Section 7 lists open questions.

2. PRELIMINARIES

2.1. Tilings. A *tiling* by a closed topological disc $X \subset \mathbb{R}^2$ is a set $\mathcal{T}$ of images gX of X under isometries g of the plane, with pairwise disjoint interiors and union $\mathbb{R}^2$. The isometries may reverse orientation. A tile gX is *unreflected* if g preserves orientation and *reflected* otherwise. A tiling is *homochiral* if all its tiles are unreflected, or all are reflected. It is *periodic* if some non-zero translation maps it to itself.

Let X be a polygon. For a tile S of a tiling and a point $x \in S$, the *angle of S at x* is the interior angle of S at x if x is a corner of S , and π if x lies on a side of S but is not a corner.

Lemma 2.1. *Let $\mathcal{T}$ be a tiling by a polygon. Every bounded set meets only finitely many tiles, and every point x has a neighbourhood that meets only the tiles containing x . The angles at x of the tiles containing x sum to 2π .*

Proof. The tiles are congruent, have disjoint interiors and contain discs of a fixed radius, so a bounded set meets finitely many of them. The tiles that do not contain x are then finitely many near x and closed, so they miss a small disc about x . In that disc, each tile containing x covers a sector of angle equal to its angle at x . The sectors have disjoint interiors and cover the disc. See also [GS87, Chapter 3]. $\square$

2.2. The polygon $\text{Tile}(1, 1)$. Let $T = \text{Tile}(1, 1)$ have vertices $v_0, \dots, v_{13}$ in anticlockwise order, with indices read modulo 14. Table 1 lists the coordinates used in [SMKG24b] and in the reference code for the Spectre, with $s = \sqrt{3}/2$. Edge e_i runs from v_i to v_{i+1} , and every edge has length 1. The edges e_9 and e_{10} are collinear, and v_{10} is a vertex with angle π .

The angle θ_i is a multiple of $\pi/2$ when i is even and a multiple of $2\pi/3$ when i is odd [SMKG24b, proof of Lemma 2.1]. We call v_i *blue* when i is even and *red* when i is odd. Thus, in anticlockwise order, e_i runs from blue to red when i is even and from red to blue when i is odd. (This parity of vertices differs from the even and odd tiles and edges of [SMKG24b, Section 3], which refer to directions.)

i	v_i	θ_i	i	v_i	θ_i
0	(0, 0)	90°	7	(3, 2)	120°
1	(1, 0)	240°	8	$(3 - s, 3/2)$	270°
2	$(3/2, -s)$	90°	9	$(5/2 - s, 3/2 + s)$	120°
3	$(3/2 + s, 1/2 - s)$	120°	10	$(3/2 - s, 3/2 + s)$	180°
4	$(3/2 + s, 3/2 - s)$	270°	11	$(1/2 - s, 3/2 + s)$	120°
5	$(5/2 + s, 3/2 - s)$	120°	12	$(-s, 3/2)$	90°
6	$(3 + s, 3/2)$	90°	13	(0, 1)	240°

TABLE 1. The vertices v_i of $\text{Tile}(1, 1)$ and its interior angles θ_i , with $s = \sqrt{3}/2$.

2.3. **Results from [SMKG24b].** We use three facts about $\text{Tile}(1, 1)$.

Theorem 2.2 (Smith, Myers, Kaplan and Goodman-Strauss).

- (1) *Every tiling by $\text{Tile}(1, 1)$ is vertex-to-vertex, with v_{10} counted as a vertex: no vertex of a tile lies in the interior of an edge of another* [SMKG24b, proof of Lemma 2.1].
- (2) *There is a homochiral tiling by $\text{Tile}(1, 1)$. Indeed, a Spectre tiles the plane* [SMKG24b, Theorem 2.2], *and every tiling by a Spectre straightens to a homochiral tiling by $\text{Tile}(1, 1)$* [SMKG24b, Section 2].
- (3) *Every homochiral tiling by $\text{Tile}(1, 1)$ is non-periodic* [SMKG24b, Section 2.1, with the proofs of Sections 3–5].

Lemma 2.3. *In a tiling by $\text{Tile}(1, 1)$, a blue vertex of one tile never coincides with a red vertex of another. So two tiles that share an edge share its blue end and its red end.*

Proof. By Theorem 2.2(1), the tiles at a vertex x of the tiling all have a vertex at x . If the blue angles there sum to $m\pi/2$ and the red ones to $2n\pi/3$, then $3m + 4n = 12$ with $m, n \geq 0$, so $m = 0$ or $n = 0$. The second claim follows, since shared edges have the same end points in both tiles. $\square$

3. THE TILES $X(p, h)$

3.1. **Construction.** Let J be the quarter turn anticlockwise. For each edge e_i define *edge coordinates* by

$$E_i(u, w) = v_i + u(v_{i+1} - v_i) + wJ(v_{i+1} - v_i).$$

Since T is anticlockwise, $w > 0$ points into T . The half turn about the midpoint of e_i acts by $(u, w) \mapsto (1 - u, -w)$.

Fix $0 < p < 1$ and $h \neq 0$. Let γ be the path $(0, 0), (p, h), (1, 0)$ in the (u, w) -plane, and $\gamma^* = \{(1 - u, -w) : (u, w) \in \gamma\}$ its half turn, the path $(0, 0), (1 - p, -h), (1, 0)$. Put

$$\gamma_i = E_i(\gamma) \quad (i \text{ even}), \quad \gamma_i = E_i(\gamma^*) \quad (i \text{ odd}),$$

let c_i be the apex of γ_i , and let Δ_i be the closed triangle $v_i c_i v_{i+1}$. The apex c_i lies at distance $|h|$ from the line through e_i , and its foot lies inside e_i . We call Δ_i a *dent* if c_i lies on the inner side of e_i , and a *bump* otherwise. For $h > 0$ the dents are the Δ_i with i even; for $h < 0$, those with i odd. Write

$$a = \sqrt{p^2 + h^2}, \quad b = \sqrt{(1 - p)^2 + h^2}, \quad \alpha = \arctan \frac{|h|}{p}, \quad \beta = \arctan \frac{|h|}{1 - p}, \quad \varphi = \pi - \alpha - \beta.$$

So φ is the angle of each Δ_i at its apex. Figure 1 shows the two paths.

We use two conditions on (p, h) .

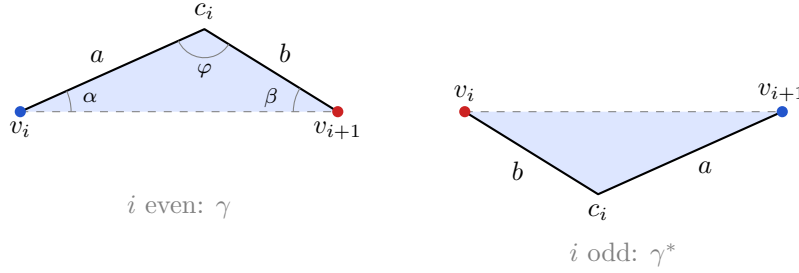


FIGURE 1. The two paths, drawn in edge coordinates with $p = 29/50$ and $h = 13/50$. The inside of the tile is above each edge. Blue marks the vertices with angles in $\{\pi/2, \pi, 3\pi/2\}$, red those with angles in $\{2\pi/3, 4\pi/3\}$. The side of length a always meets a blue vertex.

Definition 3.1.

- (G) Every dent lies in T and meets ∂T only in its base. Every bump meets T only in its base. The dents are pairwise disjoint, and so are the bumps.
- (N) There is no equation

$$\theta + m\varphi + n(2\pi - \varphi) + \sigma = 2\pi$$

with $\theta \in \{2\pi/3, 4\pi/3\}$, integers $m, n \geq 0$ with $m + n \geq 1$, and σ a sum of zero or more terms from $\{\pi/2, 2\pi/3, \pi, 4\pi/3, 3\pi/2\}$.

Condition (N) says that angle sums alone keep a red corner of $\text{Tile}(1, 1)$ away from the apexes. Section 4 shows that the side lengths do the same for most tiles where (N) fails.

Assume (G). The sides of the triangles Δ_i , other than their bases, form a closed polygonal curve $\Gamma = \gamma_0\gamma_1 \cdots \gamma_{13}$. Two triangles of the same kind lie on edges with no common vertex and are disjoint. A dent Δ_i and a bump Δ_j meet at most in $\Delta_i \cap e_j \subseteq \Delta_i \cap \partial T = e_i$, so in $e_i \cap e_j$. Hence Γ is simple. Let $X = X(p, h)$ be the closed region bounded by Γ . Then

$$X = \left(T \setminus \bigcup_{\text{dents}} (\Delta_i \setminus e_i) \right) \cup \bigcup_{\text{bumps}} \Delta_i.$$

3.2. The two conditions.

Proposition 3.2. *Let δ be the least distance between two edges of T that have no common vertex. Then $\delta = \sqrt{6 - 3\sqrt{3}}$, the distance between v_4 and v_8 , and condition (G) holds whenever $0 < |h| < \delta/2$.*

Proof. The edges of a simple polygon do not cross, so the distance between two disjoint edges is attained at an end point of one of them. A direct computation over the 77 pairs of edges with no common vertex gives $\delta^2 = |v_8 - v_4|^2 = (3/2 - \sqrt{3})^2 + 3/4 = 6 - 3\sqrt{3}$; the ancillary file checks this in exact arithmetic.

Each Δ_i lies within distance $|h|$ of e_i , since its apex is at distance $|h|$ from the line through e_i with foot inside e_i . Since $|h| < \delta$, Δ_i misses every edge that has no common vertex with e_i , and since $2|h| < \delta$, it misses every triangle on such an edge.

Now let e_j share the vertex v with e_i . The triangle Δ_i lies in the sector at v bounded by e_i and the ray from v through c_i . This sector has angle α or β , both less than $\pi/2$. It lies on the inner side of e_i for a dent and the outer side for a bump. The edge e_j makes the angle

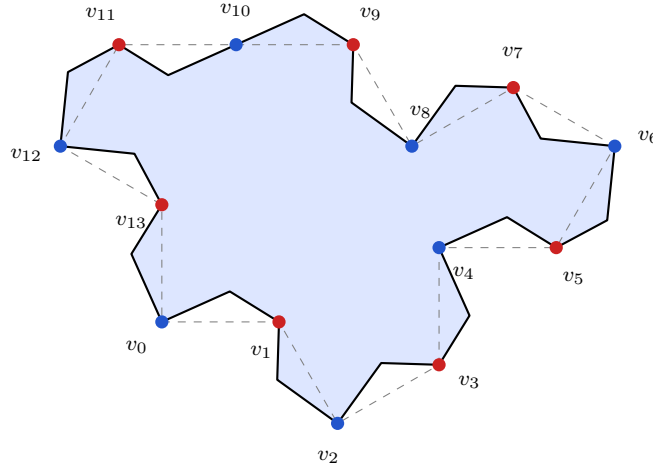


FIGURE 2. The tile $X(29/50, 13/50)$, drawn over $\text{Tile}(1,1)$ (dashed). Blue dots mark the vertices v_i with i even, red dots those with i odd. The two sides through v_{10} are collinear, so v_{10} is not a corner.

$\theta_v \geq \pi/2$ with e_i on the inner side, and $2\pi - \theta_v \geq \pi/2$ on the outer side, since $\pi/2 \leq \theta_v \leq 3\pi/2$. So $\Delta_i \cap e_j = \{v\}$.

Hence $\Delta_i \cap \partial T = e_i$. The set $\Delta_i \setminus e_i$ is connected, misses ∂T , and contains points near the midpoint of e_i on the side of c_i . So it lies in the interior of T for a dent and in the exterior for a bump. Finally, two dents lie on edges with no common vertex, since dents occur on every second edge, and so they are disjoint by the second paragraph. The same holds for bumps. $\square$

Condition (G) holds well beyond $|h| < \delta/2$: at $p = 29/50$, an exact computation finds it up to $|h| \approx 0.685$.

Lemma 3.3.

- (1) $\varphi > \pi/2 - \arctan |h|$. In particular $\varphi > \pi/3$ when $|h| < \delta/2$.
- (2) If $\varphi > \pi/3$ and (N) fails, then φ is one of $5\pi/12$, $4\pi/9$, $2\pi/3$ and $5\pi/6$. The solutions of the equation in (N) are then those in Table 2.

φ	angles at a point that sum to 2π
$5\pi/6$	$\varphi, \pi/2, 2\pi/3$
$2\pi/3$	$2\pi - \varphi, 2\pi/3; \varphi, 4\pi/3; \varphi, 2\pi/3, 2\pi/3; \varphi, \varphi, 2\pi/3$
$4\pi/9$	$\varphi, \varphi, \varphi, 2\pi/3$
$5\pi/12$	$\varphi, \varphi, \pi/2, 2\pi/3$

TABLE 2. The solutions of the equation in (N) when $\varphi > \pi/3$. Each lists θ , the apex angles and the terms of σ .

Proof. (1) The segment e_i subtends the angle φ at c_i . Let u and u' be the points at distance $|h|$ from v_i and from v_{i+1} , on the perpendiculars to e_i and on the same side as c_i . The circle through v_i, v_{i+1} and u also passes through u' , by symmetry. Since $0 < p < 1$, the apex c_i lies strictly between u and u' , so it lies inside the circle, and e_i subtends a larger angle there than at u . At u the angle is $\arctan(1/|h|) = \pi/2 - \arctan |h|$. Finally $\delta/2 < 0.45 < \tan(\pi/6)$.

(2) Since $2\pi - \varphi > \pi$ and $\theta > 0$, we have $n \leq 1$. If $n = 1$, then $(m - 1)\varphi + \theta + \sigma = 0$, so $m = 0$ and $\varphi = \theta + \sigma$. Since $\varphi < \pi$, this gives $\sigma = 0$ and $\varphi = 2\pi/3$. If $n = 0$, then $m \geq 1$ and $m\varphi = 2\pi - \theta - \sigma$. The values of $\theta + \sigma$ below 2π are $2\pi/3, 7\pi/6, 4\pi/3, 5\pi/3$ and $11\pi/6$, so $m\varphi$ is $4\pi/3, 5\pi/6, 2\pi/3, \pi/3$ or $\pi/6$. With $\pi/3 < \varphi < \pi$, the only possibilities are $\varphi = 4\pi/3 \div m$ with $m = 2, 3$, $\varphi = 5\pi/6 \div m$ with $m = 1, 2$, and $\varphi = 2\pi/3$ with $m = 1$. Reading off θ and σ in each case gives Table 2. $\square$

3.3. The corners of X .

Lemma 3.4. *Assume (G).*

- (1) *The angle of X at v_i is θ_i . In particular v_{10} is not a corner of X : the sides c_9v_{10} and $v_{10}c_{10}$ are collinear.*
- (2) *The angle of X at c_i is φ if Δ_i is a bump and $2\pi - \varphi$ if Δ_i is a dent.*
- (3) *Both sides of X at a blue vertex have length a , and both sides at a red vertex have length b .*

So X is a polygon with 27 corners: the 13 vertices v_i with $i \neq 10$, seven apexes with angle φ and seven with angle $2\pi - \varphi$. Of its 27 sides, 26 have length a or b , and the side c_9c_{10} has length $2a$ and midpoint v_{10} .

Proof. (3) For even i , γ_i runs from the blue vertex v_i to c_i with length a , then to the red vertex v_{i+1} with length b . For odd i it runs from the red vertex v_i with length b , then to the blue vertex v_{i+1} with length a .

(1) Let ρ be the rotation about v_i that maps v_{i-1} to v_{i+1} , and let R be its linear part, which commutes with J . The point $E_{i-1}(u, w)$ lies at offset $(1 - u)(v_{i-1} - v_i) - wJ(v_{i-1} - v_i)$ from v_i , so its image lies at offset $(1 - u)(v_{i+1} - v_i) - wJ(v_{i+1} - v_i)$, and

$$\rho(E_{i-1}(u, w)) = E_i(1 - u, -w).$$

Consecutive paths differ by the half turn $(u, w) \mapsto (1 - u, -w)$, so $\rho(\gamma_{i-1}) = \gamma_i$ and ρ maps the side $v_i c_{i-1}$ onto the side $v_i c_i$. At a corner of a simple anticlockwise polygon, the interior angle is the anticlockwise angle from the outgoing side to the reversed incoming side. For T at v_i these sides lie along the rays $v_i v_{i+1}$ and $v_i v_{i-1}$; for X , which is simple and anticlockwise by (G), along $v_i c_i$ and $v_i c_{i-1}$. The rotation ρ maps the second ray of each pair onto the first through the same angle. So the two interior angles are equal. At v_{10} the angle is π , so c_9, v_{10} and c_{10} are collinear.

(2) This is the apex angle of Δ_i , seen from inside a bump or from outside a dent.

The count of corners and sides follows from (1)–(3), since $|c_9v_{10}| = |v_{10}c_{10}| = a$. $\square$

Lemma 3.4(1) says that X is an instance of the second construction of [SMKG24b, Lemma 2.1], in which the paths on consecutive edges are related by the rotation about their common vertex. That lemma assumes a smooth path; here the path is a triangle.

Lemma 3.5. *Assume (G). The only isometry g with $gX = X$ is the identity.*

Proof. Number the corners of X from 0 to 26 in anticlockwise order, starting with v_0 , so that v_i has number $2i$ for $i \leq 9$ and $2i - 1$ for $i \geq 11$. A symmetry of X permutes its corners, keeps their angles, and acts on the numbers as $k \mapsto k + r$ or $k \mapsto r - k$ modulo 27. Suppose first that $\varphi \neq 2\pi/3$. Then no apex has angle $2\pi/3$ or $4\pi/3$, so the symmetry keeps the set $\{2, 25\}$ of corners with angle $4\pi/3$ and the set $\{6, 10, 14, 18, 21\}$ of corners with angle $2\pi/3$. The first set forces $r = 0$ for $k \mapsto k + r$, since $r = 23$ and $r = 4$ cannot both hold, and $r = 0$ for $k \mapsto r - k$. But $k \mapsto -k$ maps the second set to $\{21, 17, 13, 9, 6\}$. So the symmetry is the identity. If $\varphi = 2\pi/3$, then no apex has angle $\pi/2$ or $3\pi/2$, and the same argument applies to

the set $\{8, 16\}$ of corners with angle $3\pi/2$, which forces $r = 0$ or $k \mapsto 24 - k$, and to the set $\{0, 4, 12, 23\}$ of corners with angle $\pi/2$, which $k \mapsto 24 - k$ maps to $\{24, 20, 12, 1\}$. $\square$

So for each tile S of a tiling by X there is a unique isometry g with $S = gX$, and the apexes, sides and edge paths of S are the images of those of X under this g .

4. LOCAL STRUCTURE OF TILINGS BY X

Throughout this section, $X = X(p, h)$ satisfies (G), and $\mathcal{T}$ is a tiling by X . The *edge path* of a tile $S = gX$ through its apex gc_i is the image $g\gamma_i$: two sides, from one vertex gv_i to the next. A corner gv_i of S is *blue* or *red* as v_i is, and an apex gc_i is a *bump* or a *dent* as c_i is. By Lemma 3.4(3), a side of length b joins a red corner to an apex, and we call it a *b-side*. Two apexes of a tile that are next to the same vertex gv_i lie on the edges e_{i-1} and e_i , of opposite parity, so one is a bump and the other a dent.

Lemma 4.1. *At most one tile has a dent at a given point x . If one does, then exactly one other tile contains x , and its angle at x is φ .*

Proof. A dent has angle $2\pi - \varphi > \pi$, so two cannot share a point. The other tiles at x fill the angle $\varphi < \pi$. Every angle of a tile is at least $\min(\pi/2, \varphi)$, and two such angles sum to more than φ . So a single tile fills it. $\square$

We shall show that red corners meet only red corners, as in Goucher's argument. Write

(R) In every tiling by X , each point that is a red corner of some tile is a red corner of every tile containing it.

Proposition 4.2. *Condition (R) holds in each of the following cases:*

- (1) X satisfies (N);
- (2) $0 < |h| < \delta/2$.

Proof. Let x be a red corner of a tile Q . By Lemma 3.4, each tile containing x has angle π there, or the angle of a corner of $\text{Tile}(1, 1)$, or an apex angle. Suppose first that no apex lies at x . In units of $\pi/6$, the angle of Q is 4 or 8, and the other angles come from $\{3, 4, 6, 8, 9\}$ and sum to 8 or 4. The only ways are 8, 4 + 4 and 4, all red. So some tile containing x has no red corner there only if an apex lies at x , and then the angles at x solve the equation in (N). This proves (1).

(2) Condition (G) holds by Proposition 3.2. Suppose that (R) fails. Then (N) fails, so by Lemma 3.3, φ is one of the four values in Table 2, and a red corner of some tile Q lies at a point x with the angles listed there. Angle sums give four facts at every point, for these four values of φ . Recall that $\varphi > \pi/3$, so that every angle of a tile exceeds $\pi/3$.

- (F1) No apex lies in the interior of a side. An angle π plus a dent exceeds 2π , and plus a bump leaves $\pi - \varphi < 2\pi/3$. That is less than two angles, so it would be a single angle: $\pi/2$ or φ , either of which gives $\varphi = \pi/2$.
- (F2) No red corner lies in the interior of a side, since an angle π plus a red angle leaves at most $\pi/3$.
- (F3) If a blue corner lies in the interior of a side, then exactly one more tile contains the point, and it has a blue corner with angle $\pi/2$ there. The first two tiles leave exactly $\pi/2$, which one tile fills, and $\varphi \neq \pi/2$.
- (F4) An apex and a blue corner meet only as in Table 2, so only when φ is $5\pi/6$ or $5\pi/12$. The ancillary file checks this count, together with Table 2.

We use these facts through the following claim. Let S have an apex at x and lie next to Q across a ray r from x , and suppose that the side of S along r is not its b -side. Then $b = 2a$, or else $a = b$ and φ is $5\pi/6$ or $5\pi/12$.

To prove the claim, note that the side of Q along r is a b -side, ending at an apex z of Q at distance b from x . Suppose first that the side of S along r has length a , ending at a blue corner y of S . If $a > b$, then z lies in the interior of that side, against (F1). If $a = b$, then $z = y$, and (F4) applies. If $a < b$, then y lies in the interior of the side of Q . By (F3), a third tile C has a blue corner at y with angle $\pi/2$, filling the right angle between the other side of S at y and the part of r beyond y . So C has a side of length a along r , from y to an apex of C at distance $2a$ from x . Unless $2a = b$, one of this apex and z lies in the interior of a side of the other tile, against (F1). The other case is that the side of S along r is the image of c_9c_{10} , with the image of v_{10} at distance a and an apex of S at distance $2a$. If $a \geq b$, then z lies in the interior of this side, against (F1). If $a < b$, then unless $2a = b$, the far apex of S lies in the interior of the side of Q , or z lies in the interior of the side of S , against (F1). This proves the claim.

Case $\varphi = 5\pi/6$. Three tiles contain x : Q , a tile K with a blue corner of angle $\pi/2$, and a tile S with a bump. Each pair of them shares a ray from x . Along the ray shared by K and Q , the sides of K and Q end at apexes at distances a and b . If $a \neq b$, the nearer apex lies in the interior of a side of the other tile, against (F1). So $a = b$, and the two apexes meet at a point z . At z they are a bump and a dent, since two bumps leave $2\pi - 2\varphi = \pi/3$. Along the ray shared by K and S , the side of K ends at an apex at distance a . It meets a blue or red corner of S there, since the image of v_{10} is ruled out by (F1). By Table 2, this apex is a bump. So the other apex of K next to x , at z , is a dent, and the apex of Q at z is a bump. In the same way, the apex of Q on the ray shared with S meets a blue or red corner of S , so it is a bump. So both apexes of Q next to x are bumps, a contradiction.

Cases $\varphi = 5\pi/12$ and $\varphi = 4\pi/9$. Here $a \neq b$ and $b \neq 2a$. Indeed, $a = b$ means $p = 1/2$ and $|h| = \frac{1}{2} \cot(\varphi/2)$, while $b = 2a$ gives $a^2 = 1/(5 - 4\cos\varphi)$ by the law of cosines and $|h| = ab \sin\varphi = 2\sin\varphi/(5 - 4\cos\varphi)$. These heights are 0.6516 and 0.4873 for $\varphi = 5\pi/12$, and 0.5959 and 0.4575 for $\varphi = 4\pi/9$, all above $\delta/2 < 0.4483$. By Table 2, Q has angle $2\pi/3$ at x and the other tiles are bumps, apart from one tile K with a blue corner when $\varphi = 5\pi/12$. The tiles K and Q do not share a ray, by the argument of the previous case, since $a \neq b$. So both rays of Q at x are shared with tiles that have bumps at x . By the claim, their sides along these rays are b -sides, which end at red corners at the two apexes of Q next to x . By Table 2, every apex at such a point is a bump, so both apexes of Q next to x are bumps, a contradiction.

Case $\varphi = 2\pi/3$. In the first three rows for this φ in Table 2, a single tile S has an apex at x , and both of its rays are shared with tiles that have red corners at x . One of its two sides at x is not a b -side, so $b = 2a$ by the claim. In the last row, Q has angle $2\pi/3$, and both its rays are shared with tiles that have bumps at x . If the side of either along the ray is not a b -side, then $b = 2a$ by the claim. Otherwise both apexes of Q next to x lie at red corners of other tiles. If $b \neq 2a$, the same argument rules out the first three rows at those points, so the last row holds there and both apexes are bumps, a contradiction. So $b = 2a$. The law of cosines then gives $1 = a^2 + 4a^2 + 2a^2$, so $a = 1/\sqrt{7}$ and $b = 2/\sqrt{7}$. Hence $p = (1 + a^2 - b^2)/2 = 2/7$ and $|h| = ab \sin\varphi = \sqrt{3}/7$. For these two tiles, (R) holds by Proposition 4.4 below. $\square$

The tiles $X(2/7, \pm\sqrt{3}/7)$. Here $\varphi = 2\pi/3$, $a = 1/\sqrt{7}$ and $b = 2a$. The side c_9c_{10} then has the length of a b -side, and its ends have the red angles $2\pi/3$ and $4\pi/3$, so side lengths alone cannot separate these two tiles' red corners from their apexes. We cut each side of length $2a$

at its midpoint. Each tile then becomes a polygon with 42 sides of length a , which we call its *short sides*.

Lemma 4.3. *Let $X = X(2/7, \pm\sqrt{3}/7)$. In every tiling by X , each short side of a tile is a short side of exactly one other tile.*

Proof. The angles of a tile are $\pi/2$ and $3\pi/2$ at blue corners, $2\pi/3$ and $4\pi/3$ at red corners and apexes, and π inside a side. In units of $\pi/6$ these are 3, 9, 4, 8 and 6, and a sum of 12 cannot mix 4s and 8s with 3s, 6s and 9s. So a red corner or an apex never lies at a blue corner or inside a side of another tile.

Let a blue corner y of a tile C lie inside a side s of a tile F . Then the angles at y are π , $\pi/2$ and $\pi/2$, so a third tile D has a blue corner with angle $\pi/2$ at y . The sides of C and D along s have length a and end at apexes z_1 and z_2 , on either side of y . These apexes do not lie inside s , so each end w_j of s lies between y and z_j . If $w_1 \neq z_1$, then w_1 is a corner of F inside a side of C , so it is a blue corner. Then s has length a , and its other end w_2 is an apex of F strictly between y and z_2 , inside a side of D , a contradiction. So $w_j = z_j$, and s has length $2a$ with midpoint y .

Now let σ be a short side of a tile F . One end of σ is a red corner or an apex of F . By the previous two paragraphs, no corner of another tile lies inside σ . Nor does the midpoint of a side of length $2a$ of another tile. Both tiles would have angle π there, so that side would lie along σ and extend a distance a on both sides of the midpoint, so it would contain the red corner or apex at the end of σ in its interior. So a single tile lies across σ , and the short side of that tile containing σ has length a , so it is σ . $\square$

Proposition 4.4. *Condition (R) holds for $X(2/7, \sqrt{3}/7)$ and for $X(2/7, -\sqrt{3}/7)$.*

Proof. By Lemma 4.3, a tiling by X is edge-to-edge as a tiling by 42-gons. We searched exhaustively for patches around a red corner at which an apex also lies. After a reflection and a rotation of the whole tiling, the tile with that red corner is unreflected and in a fixed position. The search keeps a patch of tiles and picks a corner of the patch whose full angle is not yet covered. By Lemma 4.3, the next tile across the short side that bounds the gap shares that short side, so the search tries every tile that can lie there: 2×42 ways to place it, less those whose corner does not fit. It rejects a tile that overlaps the patch, and it rejects a patch whose tiles complete the angle at the first red corner without an apex. For both tiles, and for each of the 7 red corners, every branch is rejected before the patch covers the full angle at every corner within distance $6a$ of the red corner. The largest patch has 6 tiles, and the whole search visits fewer than 600 patches. So no tiling by X has an apex at a red corner, which is (R).

The ancillary file `search-exceptional-tile.py` performs the search in exact arithmetic in $\mathbb{Z}[\sqrt{3}]$. It confirms each rejected overlap by a second, independent test, a sampled point inside both tiles. It also checks that its tiles agree exactly with the construction of Section 3. As a control, the same search started from a red corner that meets only red corners covers the disc of radius $6a$ in every case, as it must. $\square$

Bumps fit into dents.

Lemma 4.5. *Assume (R), and let $x \in \mathbb{R}^2$.*

- (1) *If x is an apex of some tile, then exactly two tiles contain x . One has a bump at x , with angle φ , and the other a dent, with angle $2\pi - \varphi$. The two tiles have the same two boundary rays at x .*
- (2) *If x is an apex of no tile, then every tile containing x has angle $\pi/2$, $2\pi/3$, π , $4\pi/3$ or $3\pi/2$ at x . Moreover, no red corner of a tile lies in the interior of a side of another tile, and no blue corner of a tile coincides with a red corner of another.*

Proof. (1) Let S be a tile with an apex at x , and let w be the red corner of S at the other end of its b -side at x . By (R), every tile containing w has a red corner there, with angle $2\pi/3$ or $4\pi/3$, so two or three tiles contain w . Consecutive tiles around w share a boundary ray, and each has a b -side along it, ending at an apex. So the tile S' next to S across the ray from w through x has a b -side along wx and an apex at x . We call S' the *partner* of S at x . Near x the two tiles lie on opposite sides of wx , so S is also the partner of S' .

We show that S and S' have apexes of opposite kinds at x . By Lemma 4.1, they are not both dents. Suppose that both are bumps. Then the other apexes z of S and z' of S' next to w are dents. If only S and S' contain w , they share both rays at w , and $z = z'$ is a dent of both, against Lemma 4.1. Otherwise a third tile U contains w , with b -sides along wz and wz' , so U has apexes at z and z' . By Lemma 4.1, the angle of U at z is φ , so U has a bump at z , and likewise at z' . But these are the two apexes of U next to its red corner w , so they are not both bumps.

So the partner relation matches the tiles with a bump at x to the tiles with a dent at x . By Lemma 4.1, at most one tile has a dent at x . So exactly one tile has a bump at x and exactly one has a dent. Their angles sum to 2π , so no other tile contains x , and the two sectors share their boundary rays.

(2) The first claim is Lemma 3.4. By part (1), no corner of a tile other than an apex lies at an apex of another tile, so the remaining claims concern points that are apexes of no tile. An angle π plus a red angle leaves at most $\pi/3$, which is less than every available angle. A blue angle ($\pi/2$ or $3\pi/2$) plus a red angle ($2\pi/3$ or $4\pi/3$) leaves $5\pi/6$, $\pi/6$ or a negative amount, and neither $5\pi/6$ nor $\pi/6$ is a sum of available angles. $\square$

Lemma 4.6. *Assume (R). Let x be an apex of a tile A , and let B be the other tile containing x . Then A and B have the same hand, and their edge paths through x coincide, end points included.*

Proof. By Lemma 4.5(1), A and B share both boundary rays at x . Along each ray, each tile has a maximal straight piece of boundary starting at x , which we call its *run*. By Lemma 3.4, a run is one of:

- a side of length b , ending at a red corner (a *b-run*);
- a side of length a , ending at a blue corner; or the side c_9c_{10} , of length $2a$, which passes v_{10} at distance a and ends at an apex (both *a-runs*).

Walk around each tile anticlockwise. On an unreflected tile, by Lemma 3.4(3), the walk passes γ_i as a -side then b -side when i is even, and as b -side then a -side when i is odd. Dents occur on the edges of one parity and bumps on the other, so dents and bumps are passed in opposite orders. A reflection reverses the order in which every edge path is passed and keeps dents as dents. So on a reflected tile, dents and bumps are also passed in opposite orders, each reversed.

At x each tile's sector lies to the left of its own walk. The two sectors are complementary, so the two walks pass the two rays in opposite orders. One of A and B has a dent at x and the other a bump.

Same hand. The edge paths are passed in opposite orders, and the walks run in opposite directions, so the two runs on each ray are of the same kind. The two b -runs end together at a red corner of each tile. The two a -runs both reach distance a at a blue corner or at v_{10} of each tile. So the two edge paths through x are the same set, and their end points are images of vertices of T in both tiles.

Opposite hands. Now the edge paths are passed in the same order, so each ray carries one a -run and one b -run. Fix a ray r , let N be the tile with the b -run on r and M the tile with

the a -run, and let y be the red corner of N at distance b from x along r (Figure 3). We derive a contradiction in each case.

Case $a > b$. Then y lies in the interior of M 's run, a single side of M . This contradicts Lemma 4.5(2).

Case $a = b$. If M 's run is the side c_9c_{10} , then y lies in its interior. Otherwise M 's run ends at a blue corner of M at y . Both contradict Lemma 4.5(2).

Case $a < b$. Suppose first that M 's run is the side c_9c_{10} , ending at an apex z of M at distance $2a$ from x . If $2a < b$, then z lies in the interior of a side of N , against Lemma 4.5(1). If $2a > b$, then y lies in the interior of a side of M , against Lemma 4.5(2). If $2a = b$, then an apex and a red corner coincide, against Lemma 4.5(1).

Otherwise M 's run ends at a blue corner y' of M at distance $a < b$, in the interior of N 's side. The point y' is an apex of no tile, by Lemma 4.5(1). At y' , N has angle π . The angle of M at its blue corner y' is $\pi/2$ or $3\pi/2$, and at most π , so it is $\pi/2$. By Lemma 4.5(2), a single further tile C fills the remaining right angle, with a corner of angle $\pi/2$. That corner is a vertex of C with angle $\pi/2$, so it is blue. The sector of C at y' lies between the other side of M at y' and the continuation of r beyond y' . So C has a side along r , starting at y' and running away from x . By Lemma 3.4(3) it has length a , and since the angle of C at y' is not π , it ends at an apex z of C at distance $2a$ from x . Comparing $2a$ with b as in the previous paragraph, with C in place of M , gives a contradiction in each case. $\square$

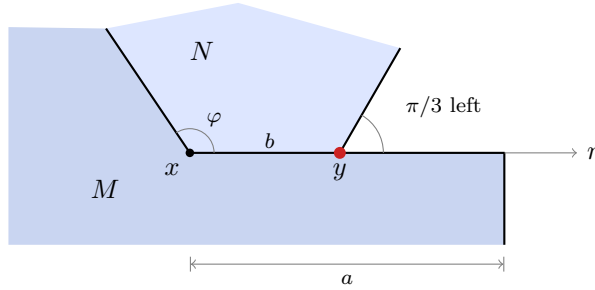


FIGURE 3. Two tiles of opposite hands at an apex x , in the case $a > b$. Here N has a bump apex at x and M a dent apex. Along the ray r , the b -run of N ends at a red corner y inside the a -run of M . The angle left at y is at most $\pi/3$, and no tile has so small an angle.

5. PROOF OF THE MAIN THEOREM

Theorem 5.1. *Let $0 < p < 1$ and $h \neq 0$ satisfy (G) and (R). Then $X = X(p, h)$ is a strictly chiral aperiodic monotile:*

- (1) X tiles the plane;
- (2) every tiling by X is homochiral;
- (3) no tiling by X is periodic.

Moreover, X is a Spectre in the sense of [SMKG24b, Section 2].

By Proposition 4.2(1), the theorem holds whenever (G) and (N) hold. Theorem 1.1 follows from Theorem 5.1, Propositions 3.2 and 4.2(2), and Lemma 3.4.

Proof. Fix a tiling $\mathcal{T}$ by X .

Step 1: edge paths match. Every edge path of every tile contains an apex. By Lemmas 4.5(1) and 4.6, each edge path of a tile S coincides with an edge path of exactly one other tile, which we call the *partner* of S along that path. The partner has the same hand as S . The triangle between the shared path and its chord is a bump of one of the two tiles and a dent of the other.

Step 2: one hand. Let $x \in \mathbb{R}^2$, and list the tiles containing x in cyclic order around x . Two consecutive tiles share a short boundary segment starting at x . It lies in an edge path of the first tile, whose partner lies across the segment and is therefore the second tile. So consecutive tiles have the same hand, and all tiles containing x have the same hand. By Lemma 2.1, a small disc about x meets only these tiles. So the points covered by unreflected tiles form an open set, and so do the points covered by reflected tiles. These sets are disjoint and cover the connected plane, so one of them is empty. This proves (2).

Step 3: straightening. For a tile $S \in \mathcal{T}$, let $\hat{S} = gT$, where g is the isometry with $S = gX$, unique by Lemma 3.5. The polygon $\hat{S}$ is S with each bump removed and each dent filled. By Step 1, each such triangle is a bump of one tile and a dent of its partner, so straightening moves it from the one to the other. Each triangle lies in the tile for which it is a bump, and the bumps of one tile are disjoint by (G), so the triangles have disjoint interiors. Hence $\hat{\mathcal{T}} = \{\hat{S} : S \in \mathcal{T}\}$ is a tiling by T , and it is homochiral by Step 2.

Step 4: no period. The map $S \mapsto \hat{S}$ commutes with isometries of the plane. If $\mathcal{T} + v = \mathcal{T}$, then $\hat{\mathcal{T}} + v = \hat{\mathcal{T}}$, and Theorem 2.2(3) gives $v = 0$. This proves (3).

Step 5: a tiling exists. Let $\{g_k T\}$ be a tiling by unreflected copies of T , which exists by Theorem 2.2(2). Two tiles $g_k T$ and $g_l T$ that share an edge run along it in opposite directions, and by Lemma 2.3 they agree on which end is blue. So the edge is $g_k e_i$ with i even in one tile, running from blue to red, and $g_l e_j$ with j odd in the other. The point $g_k E_i(u, w)$ equals $g_l E_j(1 - u, -w)$. Since γ^* is the half turn of γ , both tiles place the same path along the edge, with one triangle as a bump of one tile and a dent of the other. By (G), the dents of each $g_k X$ lie in $g_k T$ and are pairwise disjoint. So the sets $g_k X$ have disjoint interiors and cover the plane. This proves (1).

Spectre. The definition in [SMKG24b, Section 2] asks for four properties of X : every tiling by X is homochiral (Step 2); if $\{g_i X\}$ is a tiling then so is $\{g_i T\}$ (Step 3); if $\{g_i T\}$ is a homochiral tiling then so is $\{g_i X\}$ (Step 5, applied to either hand); and the two tilings have the same points that lie in three or more tiles. For the last, straightening changes the tiles containing a point only inside the moved triangles. Apart from its base end points, each point of a triangle lies in at most two tiles, before and after. The base end points are vertices of both tiles sharing the edge, before and after. $\square$

6. EXAMPLES

Proof of Corollary 1.2. Both parts follow from Theorem 1.1, since $|h| < \delta/2$. In (1), $|h| = 0.26 < 0.448$ and $\varphi \approx 124.095^\circ$, which is none of the four values in Lemma 3.3, so (N) already holds. In (2), Goucher's polygon has $\varphi = 2\pi/3$ and $a = b$. There (N) fails, and the case $\varphi = 2\pi/3$ of Proposition 4.2 applies. $\square$

The tile $X(29/50, 13/50)$ is the *Triangle* edge shape, in the *Single* arrangement, of the author's interactive tiling viewer at <https://drbuild.uk/tiles>, with peak position 0.58 and height 0.26. Figure 4 shows a patch of its tiling, built from a homochiral tiling by $\text{Tile}(1, 1)$ generated by the substitution rules of [SMKG24b, Figure 2.1 and Appendix A].

Computer check. Apart from Proposition 4.4, the proofs above need no computer. As a check, the ancillary file `verify-polygonal-spectre.py` recomputes, in exact arithmetic over $\mathbb{Q}(\sqrt{3})$, the angles of $\text{Tile}(1, 1)$ and their colours, the value of δ , Lemma 3.4(1) and (3),

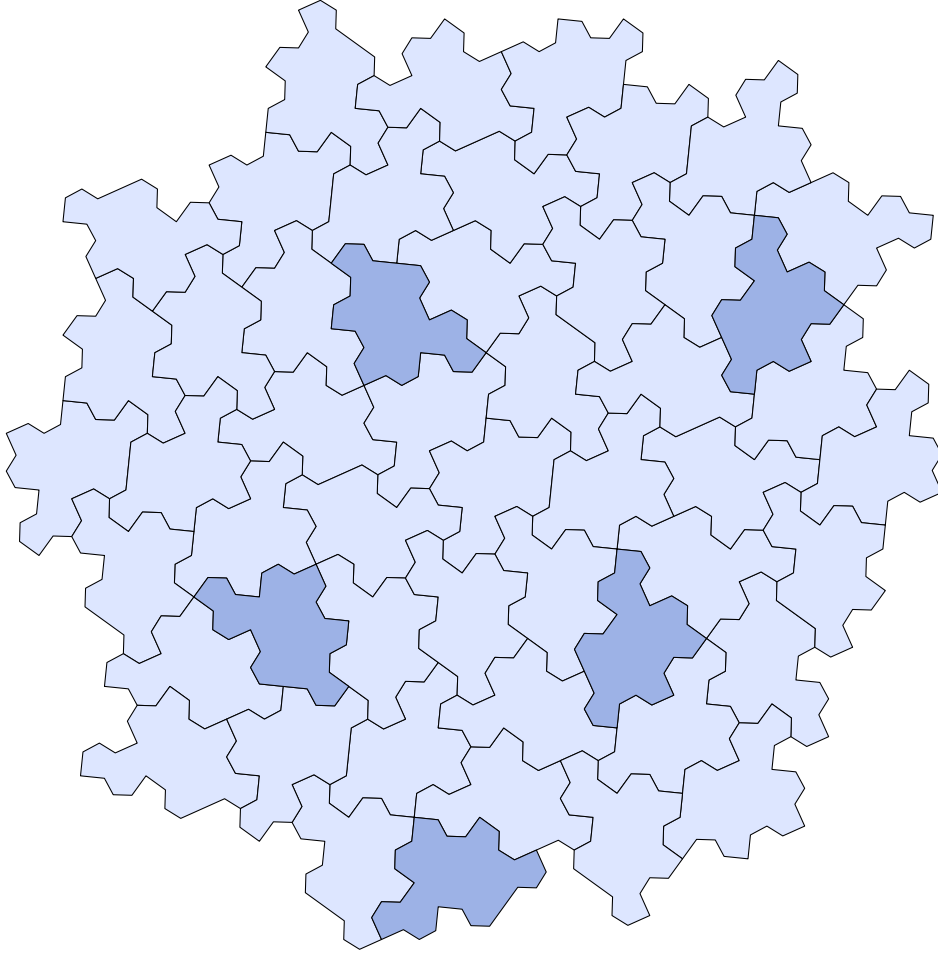


FIGURE 4. A patch of a tiling by $X(29/50, 13/50)$. Darker tiles are turned by an odd multiple of $\pi/6$ relative to the lighter ones.

condition (G) for any rational (p, h) , Table 2 with fact (F4), and Lemma 3.5. It compares the four heights in the proof of Proposition 4.2 with $\delta/2$ in floating point, where the margin exceeds 0.009, and uses no floating point in any other test. The second ancillary file, `search-exceptional-tile.py`, carries out the search in the proof of Proposition 4.4.

7. REMARKS AND OPEN QUESTIONS

Remark 7.1. Goucher’s 28-edge polygon [Gou23a, Gou23b] has unit sides meeting at turns of $\pm\pi/3$ at the apexes, so it is $X(1/2, \sqrt{3}/6)$ scaled by $\sqrt{3}$. Its apex angles are $2\pi/3$ and $4\pi/3$, the same as the red angles of $\text{Tile}(1, 1)$, so angle sums cannot tell an apex from a red corner, and (N) fails. Goucher’s argument [Gou23b] tells them apart by position. Angle sums show that blue corners meet only blue corners. A red corner lies two sides from a blue corner and an apex lies one side from one, so red corners meet only red corners. Each apex then meets only an apex with the complementary angle, and this forces a single hand. Our proof follows the same plan. Proposition 4.2 shows that red corners meet only red corners, and Lemma 4.5(1) pairs each apex with the apex at the far end of the b -side from a shared red corner.

Remark 7.2. In the range $0 < |h| < \delta/2$, only the tiles $X(2/7, \pm\sqrt{3}/7)$ need the computer (Proposition 4.4). The search trees are small, so a proof by hand should exist, but we have not written one.

Remark 7.3. For $|h| \geq \delta/2$, Theorem 5.1 applies wherever (G) and (N) hold. Condition (N) then fails at more values of φ , such as $\varphi \leq \pi/3$, and we have not examined them.

Question 7.4. Does every non-straight piecewise-linear path, placed on the edges of $\text{Tile}(1, 1)$ by either construction of [SMKG24b, Lemma 2.1] so that the tile is a topological disc, give a Spectre? In particular, is the half-turn-symmetric zigzag $(0, 0), (q, k), (1 - q, -k), (1, 0)$ a Spectre for small $k \neq 0$?

Question 7.5. What is the least number of sides of a polygon that is a strictly chiral aperiodic monotile? The tiles here have 27 sides.

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